

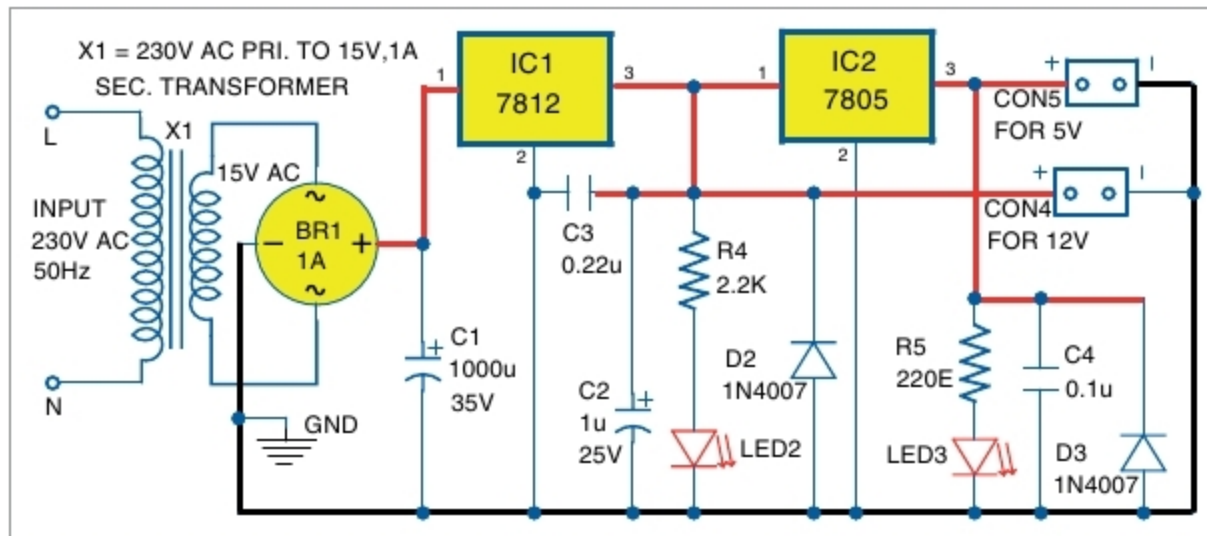


Fig. 3: Circuit diagram of power supply

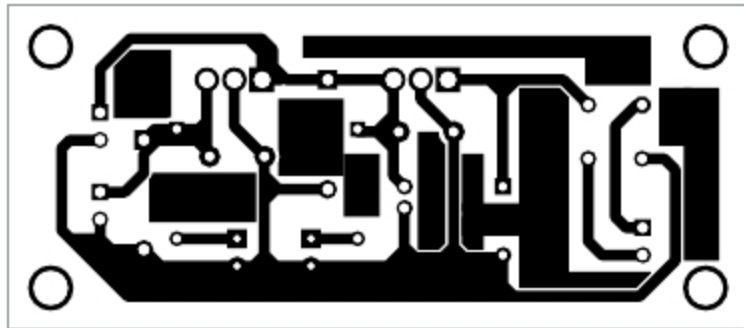


Fig. 4: Actual-size PCB layout for power supply

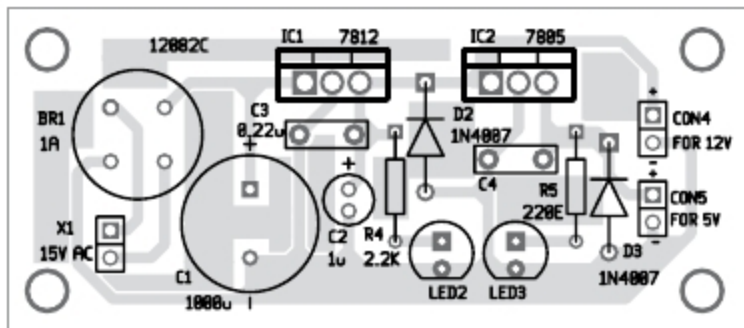


Fig. 5: Components layout of the PCB

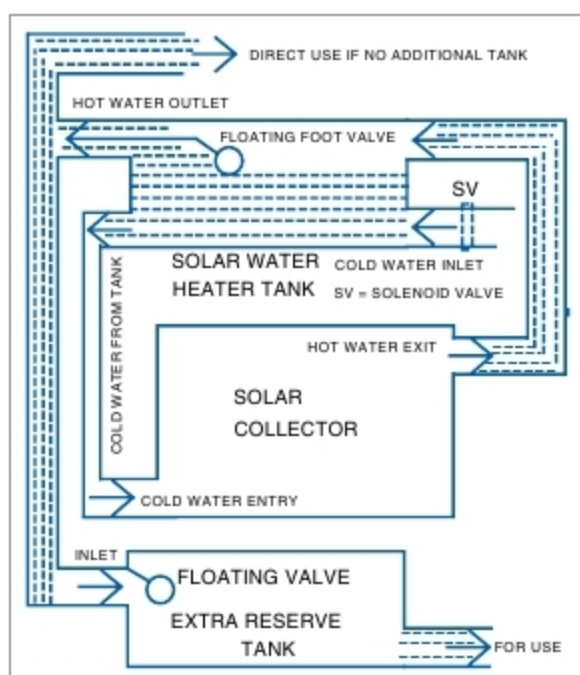


Fig. 6: Block diagram of reserve tank with floating valve

light detector. The resistance of LDR1 decreases with the increase in light intensity bombarded on it. In absence of light, the resistance of LDR1 may reach up to a few mega-ohms (MΩ), while in the presence of light, its

resistance reduces to a few ohms (Ω). The reduction in its internal resistance increases conductivity.

Relay. Relay RL1 is an electromagnetic switching device, which provides complete electrical isolation between the controlling circuit and output circuit. It is controlled/triggered by a low DC. A 12V single-changeover relay is used here to switch the solenoid valve on and off with the use of an electrical pulse.

Solenoid valve. Solenoid valve (SV) is an electrically operated valve. Basically, there are two types of solenoid valves according to the operation state: normally open (NO) and normally closed (NC). These valves are used to control the flow of fluid. If the flow of fluid is stopped when the solenoid valve is energised, it is a normally open (NO) solenoid valve. If the flow of fluid is stopped when the solenoid is de-energised, it is a normally closed (NC) solenoid valve. Here, a 12V DC NC solenoid valve is used. The water flows when the solenoid is energised.

Software. Arduino program (Utilization_enhancer.ino) is used for this prototype. This program code must be uploaded to Arduino Uno using Arduino IDE. In the program, digital pin 13 of the Arduino board is defined as the output. Its analogue pin A0 is used to read analogue input signals. Some main functions of the Arduino code are explained below:

pinMode(). It configures the specified pin to behave either as an input or output. See the description of digital

pins for the functionality of the pins.

DigitalWrite(). If any Arduino pin is configured as output with pinMode(), its voltage will be set to corresponding values: 5V (or 3.3V on 3.3V board) for high and 0V (ground) for low.

Serial. It is used for communication between the Arduino board and computer or other devices. All Arduino boards have at least one serial port (also known as UART or USART).

Begin(). It sets the data rate in bits per second (baud) for serial data transmission. For communicating with the computer, use one of these rates: 300, 600, 1200, 2400, 4800, 9600, 14400, 19200, 28800, 38400, 57600, or 115200. You can, however, specify other rates—for example, you can communicate over pins 0 and 1 with a component that requires a particular baud rate.

AnalogRead(). It reads the value from the specified analogue pin. The Arduino board contains a multichannel, 10-bit analogue-to-digital converter. This means that it will map input voltages between 0 and operating voltage (5V or 3.3V) into integer values between 0 and 1023.

println(). It prints data on the serial port as human-readable ASCII text followed by a carriage return character.

Delay(). It is used to pause the program for the amount of time (in milliseconds) specified as a parameter.

Construction and testing

First, assemble the power supply circuit. An actual-size PCB layout for the power supply section is shown in Fig. 4 and its components layout in Fig. 5. After assembling the circuit on the PCB, connect it with the Arduino Uno board, relay, and solenoid valve as shown in Fig. 1 via jumper/external wires.

Before assembling the main circuit (Fig. 2) on a breadboard or any general-purpose PCB or veroboard, do not forget to upload the Arduino program code into the Arduino board.

As shown in the main circuit in Fig. 2, a photoresistor (LDR1) is connected between analogue (A0) and +5V pins of the Arduino Uno board. An NPN switching transistor 2N2222 (T1) is used to drive the